

Information theory, self-organisation and complex systems

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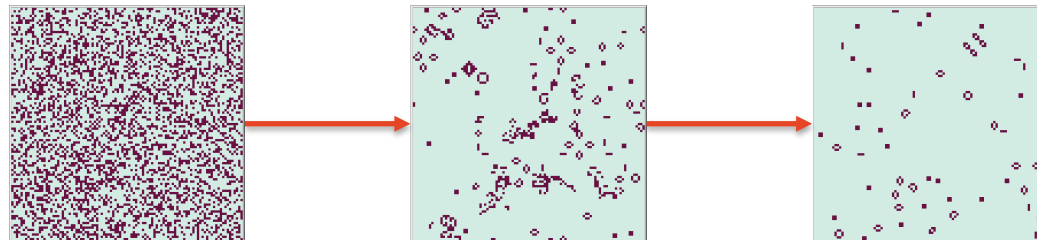


Self-organisation: session outcomes

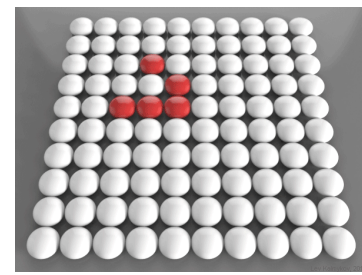
- Understand concept of self-organisation;
 - Understand candidate approaches to measure it directly at system level, and within a system.
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- Primary references:
 - J.T. Lizier, "The local information dynamics of distributed computation in complex systems", Springer, Berlin/Heidelberg, 2013. Section 2.1.2 ([pre-print](#))
 - M. Prokopenko, F. Boschetti, A.J. Ryan, "An Information-Theoretic Primer on Complexity, Self-Organization, and Emergence", Complexity, 15(1), pp. 11-28, 2009.

Self-organisation

- Game of Life – a canonical complex system
- Run it:
 - In NetLogo (Models library | Sample Models | Computer Science | Cellular Automata | Life), or
 - Directly on the [NetLogo website](#)



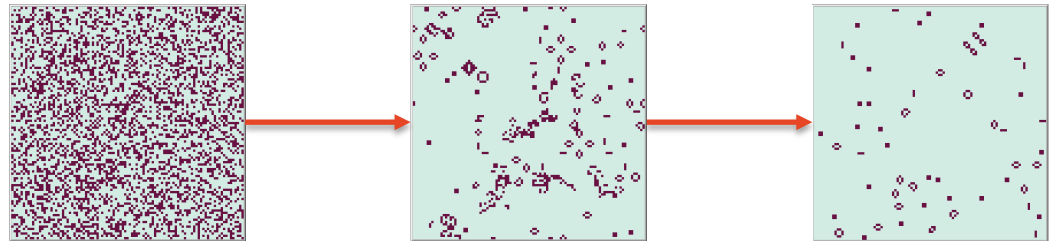
- From your observations:
 - Describe the intermediate and final states in contrast to the initial random state?
 - Did it organise? In what way?
 - How did that happen? Was there any central control?
 - What happens to the density, and does the original density matter?
- Other examples of self-organisation?



Glider

(Image By Lev Kalmykov (Own work) [CC BY-SA], via Wikimedia Commons)

Self-organisation



- Sayama, p. 6: *“Self-organization is a dynamical process by which a system spontaneously forms nontrivial macroscopic structures and/or behaviors over time.”*
- More specifically [2,3 in 4,5], must have 2 key features:
 - “An increase in **organisation** over **time**”
 - “Dynamics not guided by any centralised or external control agent”

[1] H. Sayama, *“Introduction to the Modeling and Analysis of Complex Systems”*, Geneseo, NY: Open SUNY Textbooks, 2015; chapter 1

[2] C.R. Shalizi, K.L. Shalizi, R. Haslinger, *“Quantifying self-organization with optimal predictors”*, Phys. Rev. Lett. 93(11), 118701 (2004)

[3] D. Polani, *“Foundations and formalizations of self-organization”*, in *Advances in Applied Self-organizing Systems*, ser. Advanced Information and Knowledge Processing, ed. by M. Prokopenko (Springer, London, 2008), pp. 19–37

[4] J.T. Lizier, *“The local information dynamics of distributed computation in complex systems”*, Springer: Berlin/Heidelberg, 2013

[5] M. Prokopenko, F. Boschetti, A.J. Ryan, *“An Information-Theoretic Primer on Complexity, Self-Organization, & Emergence”*, Complexity, 15(1), pp. 11-28, 2009.

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How to directly measure order/organisation?

Several options:

1. Complement of randomness / entropy [1]
2. Mutual information between parts of the system [2]
 - Integration/multi-information:

$$I(X_1; \dots; X_k) = \left(\sum_{i=1}^k H(X_i) \right) - H(X_1, \dots, X_k)$$

- Implemented in JIDT

`(infodynamics.measures.*.*.MultiInfoCalculator[TYPE])`

[1] C. Gershenson, N. Fernández, "Complexity and information: Measuring emergence, self-organization, and homeostasis at multiple scales", *Complexity*, 18(2), pp. 29-44, 2012

[2] D. Polani, "Foundations and formalizations of self-organization", in *Advances in Applied Self-organizing Systems*, ser. Advanced Information and Knowledge Processing, ed. by M. Prokopenko (Springer, London, 2008), pp. 19–37

How to directly measure order/organisation?

Several options:

1. Complement of randomness / entropy [1]
 2. Mutual information between parts of the system [2]
 3. Statistical complexity* [3]
- I prefer approach 3 as a measure of organisation (*subjective*);
 - Approach 2 is more accessible to the reader and to compute.

[1] C. Gershenson, N. Fernández, "Complexity and information: Measuring emergence, self-organization, and homeostasis at multiple scales", *Complexity*, 18(2), pp. 29-44, 2012

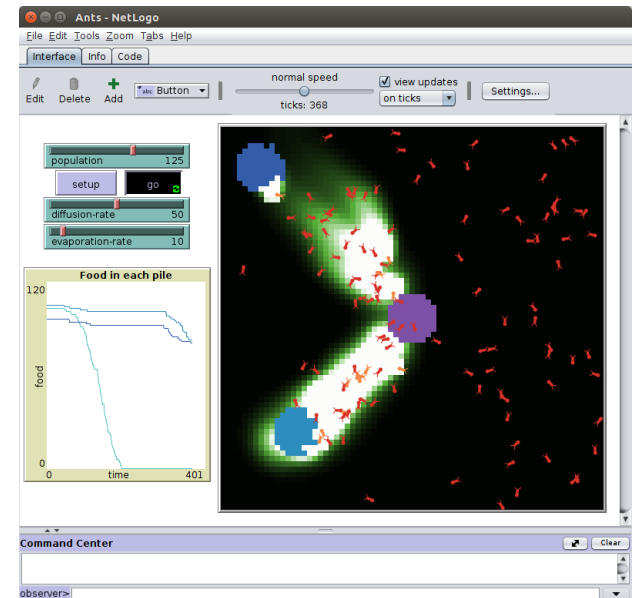
[2] D. Polani, "Foundations and formalizations of self-organization", in *Advances in Applied Self-organizing Systems*, ser. Advanced Information and Knowledge Processing, ed. by M. Prokopenko (Springer, London, 2008), pp. 19–37

[3] C.R. Shalizi, K.L. Shalizi, R. Haslinger, "Quantifying self-organization with optimal predictors", *Phys. Rev. Lett.* 93(11), 118701 (2004)

* Not covered in this course!

Alternative: measure various aspects of order/organisation

- Self-organisation is about *information structuring* (see Sayama).
- May be multiple aspects of information structure/organisation that we wish to investigate, e.g.
 - Temporal structure
 - Relationships between variables
 - Spatial structure
 - Information storage
 - Information transfer
 - Aspects that correlate with task
- Our approach:
 - To use information theory to characterise information processing structure in complex systems and their self-organisation, and how this changes over time.



Self-organisation: summary

- Self-organisation is an increase in order over time (without external control).
 - The key to measuring it is measuring order/organisation/structure in the system;
 - There are several options for doing so:
 - Directly, at system level; or
 - Examining multiple aspects of information structuring.

- *Coming up:* Information processing in complex systems.

Questions



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